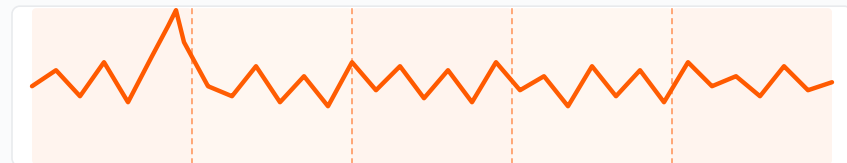


A. Signal Tokenizer

Raw vibration signal



patch (L = 4)

Transformer Encoder

4 layers, d = 128

self-attention over full signal

FSQ [5, 5, 5, 5]

<ts_412> <ts_87> <ts_553> <ts_19> ...

625 codes, 100% utilization
frozen after pretraining

compression
4:1

B. Vocabulary Extension

LLM token embedding table

"bearing"

"fault"

"vibration"

"the"

...

Original vocab (32K)

<ts_0> ... <ts_624>

+ 625 signal tokens

New (signal)

<ts_start>

<ts_end>

shared embedding space

extension
vocab

C. LLM Reasoning

Unified input sequence

[User]

Signal:

mean=0.12

std=0.87

scale context

<ts_start>

<ts_412>

<ts_87>

<ts_553>

<ts_19>

...

<ts_end>

signal tokens

What

fault

is

present?

[Asst]

task prompt

Standard self-attention

no cross-attention, no gating

OUTPUT

"Outer race bearing fault.

The 162 Hz peak with shaft-speed sidebands
is consistent with BPFI."

DoRA adapters + 627 new embeddings (<1% params)